

Stat 517 Summer 2026
Homework # 4

1. Let $Y \sim \text{Uniform}[0, 1]$ and $X_n = Y^n$. Prove X_n converges in probability to zero.

Solution: $\forall 0 < \epsilon < 1$,

$$\begin{aligned} P(|X_n - 0| \geq \epsilon) &= P(Y^n \geq \epsilon) \\ &= P(Y \geq \epsilon^{1/n}) \\ &= 1 - \epsilon^{1/n} \\ &\rightarrow 0, \end{aligned}$$

as $n \rightarrow \infty$. Therefore, X_n converges to zero in probability.

2. Let the random variable Y_n have a distribution that is $b(n, p)$.

- (a) Prove that Y_n/n converges in probability to p .
- (b) Prove that $1 - Y_n/n$ converges in probability to $1 - p$.
- (c) Prove that $(Y_n/n)(1 - Y_n/n)$ converges in probability to $p(1 - p)$.

Solution:

- (a) Because $E(Y_n/n) = p$ and $V(Y_n/n) = p(1 - p)/n$, by Chebyshev inequality, for any $\epsilon > 0$ we have

$$P(|\frac{Y_n}{n} - p| \geq \epsilon) \leq \frac{V(Y_n/n)}{\epsilon^2} = \frac{p(1 - p)}{n\epsilon^2} \Rightarrow 0,$$

as $n \rightarrow \infty$. Therefore,

$$\frac{Y_n}{n} \xrightarrow{P} p.$$

- (b) Let $g(t) = 1 - t$. By the continuous mapping theorem, we have

$$g(\frac{Y_n}{n}) \xrightarrow{P} g(p) \Rightarrow 1 - \frac{Y_n}{n} \xrightarrow{P} 1 - p.$$

- (c) Let $g(t) = t(1 - t)$. By the continuous mapping theorem, we have

$$g(\frac{Y_n}{n}) \xrightarrow{P} g(p) \Rightarrow \frac{Y_n}{n}(1 - \frac{Y_n}{n}) \xrightarrow{P} p(1 - p).$$

3. Let W_n denote a random variable with mean μ and variance b/n^p , where $p > 0$, μ and b are constants (not function of n). Prove that W_n converges in probability to μ .

Solution: By Chebyshev Inequality, we have

$$P(|W_n - \mu| \geq \epsilon) \leq \frac{V(W_n)}{\epsilon^2} = \frac{b}{n^p \epsilon^2} \rightarrow 0$$

as $n \rightarrow \infty$. Therefore,

$$W_n \xrightarrow{P} \mu.$$

4. Let $X_1, \dots, X_n$ be iid random variables with common pdf

$$f(x) = \begin{cases} e^{-(x-\theta)} & x > \theta, -\infty < \theta < \infty \\ 0 & \text{elsewhere.} \end{cases}$$

This pdf is called the shifted exponential. Let $Y_n = \min \{X_1, \dots, X_n\}$. Prove that $Y_n \rightarrow \theta$ in probability by first obtaining the cdf of Y_n .

Solution: Since Y_n is the minimum among the random variables, we can consider the following probability:

$$\begin{aligned} P(Y_n \geq y) &= P(X_1 \geq y, \dots, X_n \geq y), \\ &= \prod_{i=1}^n P(X_i \geq y), \\ &= \prod_{i=1}^n (1 - P(X_i \leq y)), \\ &= \prod_{i=1}^n \left(1 - \int_{\theta}^y e^{-(x-\theta)} dx\right), \quad y > \theta \\ &= \prod_{i=1}^n e^{-(y-\theta)}, \\ &= e^{-n(y-\theta)}, \end{aligned}$$

which implies

$$F_{Y_n}(y) = P(Y_n \leq y) = 1 - e^{-n(y-\theta)}, \quad y > \theta.$$

Let $Z_n = Y_n - \theta$, then the cdf of Z_n is

$$F_{Z_n}(z) = 1 - e^{-nz}, \quad z > 0.$$

For some $\epsilon > 0$,

$$\begin{aligned} P(|Z_n| \geq \epsilon) &= P(Z_n \geq \epsilon) \\ &= P(Y_n - \theta \geq \epsilon) \\ &= e^{-n\epsilon}, \end{aligned}$$

which converges to 0 as $n \rightarrow \infty$. Hence the fact that $Z_n \rightarrow 0$ in probability means $Y_n \rightarrow \theta$ in probability.

5. Let Y_1 denote the minimum of a random sample size n from a distribution that has pdf $f(x) = e^{-(x-\theta)}, \theta < x < \infty$, zero elsewhere. Let $Z_n = n(Y_1 - \theta)$. Investigate the limiting distribution of Z_n .

Solution: From the previous homework problem, we can say that Y_1 has the following cdf:

$$F_{Y_1}(y) = 1 - e^{-n(y-\theta)}, \quad y > \theta,$$

which implies

$$\begin{aligned} F_{Z_n}(z) &= P(Z_n \leq z) \\ &= P(n(Y_1 - \theta) \leq z) \\ &= P(Y_1 \leq z/n + \theta) \\ &= 1 - e^{-z}, \quad z > 0. \end{aligned}$$

Hence we can conclude that Z_n has an exponential distribution with parameter 1 regardless of the sample size n .

6. Let Y_n denote the maximum of a random sample of size n from a distribution of the continuous type that has cdf $F(x)$ and pdf $f(x) = F'(x)$. Find the limiting distribution of $Z_n = n[1 - F(Y_n)]$.

Solution: Let $X_1, \dots, X_n$ be a random sample from $F(x)$, then $Y_n = \max\{X_1, \dots, X_n\}$. First, let's find the cdf of Y_n .

$$\begin{aligned} P(Y_n \leq y) &= P(X_1 \leq y, \dots, X_n \leq y) \\ &= \prod_{i=1}^n P(X_i \leq y) \\ &= F(y)^n. \end{aligned}$$

Now define $U = F(Y_n)$, then the cdf of U is

$$\begin{aligned} P(U \leq u) &= P(F(Y_n) \leq u) \\ &= P(Y_n \leq F^{-1}(u)) \\ &= F(F^{-1}(u))^n \\ &= u^n. \end{aligned}$$

So the cdf of Z_n is

$$\begin{aligned} P(Z_n \leq z) &= P(n[1 - F(Y_n)] \leq z) \\ &= P\left(F(Y_n) \geq 1 - \frac{z}{n}\right) \\ &= 1 - P\left(F(Y_n) \leq 1 - \frac{z}{n}\right) \\ &= 1 - \left(1 - \frac{z}{n}\right)^n. \end{aligned}$$

Hence,

$$P(Z_n \leq z) \rightarrow 1 - e^{-z},$$

as $n \rightarrow \infty$. We can say that the limiting distribution of Z_n has an exponential distribution with a parameter 1.

7. Let Y be $b(72, 1/3)$. Approximate $P(22 \leq Y \leq 28)$.

Solution: For $Y \sim b(n, p)$ with $n = 72$, $p = 1/3$,

$$E(Y) = np = 72 \cdot \frac{1}{3} = 24, \quad \text{Var}(Y) = np(1 - p) = 72 \cdot \frac{1}{3} \cdot \frac{2}{3} = 16,$$

so the standard deviation is $\sqrt{16} = 4$. By the central limit theorem,

$$Y \stackrel{\sim}{\sim} N(24, 4^2).$$

Normal approximation.

$$\begin{aligned} P(22 \leq Y \leq 28) &= P\left(\frac{22 - 24}{4} \leq \frac{Y - 24}{4} \leq \frac{28 - 24}{4}\right) = P(-0.5 \leq Z \leq 1) \\ &= \Phi(1) - \Phi(-0.5) = 0.8413 - 0.3085 = 0.5328, \end{aligned}$$

where $\Phi(\cdot)$ is the standard normal cdf.

With continuity correction. Since Y is discrete (integer-valued) while the normal is continuous, we widen each endpoint by 0.5:

$$\begin{aligned} P(22 \leq Y \leq 28) &\approx P(21.5 \leq Y^* \leq 28.5) = P\left(\frac{21.5 - 24}{4} \leq Z \leq \frac{28.5 - 24}{4}\right) \\ &= P(-0.625 \leq Z \leq 1.125) = \Phi(1.125) - \Phi(-0.625) \approx 0.6037. \end{aligned}$$

For comparison, the exact binomial probability is 0.6001, so the continuity-corrected approximation (0.6037) is considerably closer than the uncorrected one (0.5328).

8. Let $f(x) = 1/x^2$, $1 < x < \infty$, zero elsewhere, be the pdf of a random variable X . Consider a random sample of size 72 from this distribution. Compute approximately the probability that more than 50 of the observations are less than 3.

Solution: **Step 1: Probability that a single observation is less than 3.**

$$P(X < 3) = \int_1^3 \frac{1}{x^2} dx = \left[-\frac{1}{x} \right]_1^3 = -\frac{1}{3} - (-1) = \frac{2}{3}.$$

Step 2: Count of observations less than 3. Let Y be the number of the 72 observations that are less than 3. Each observation independently satisfies this with probability $p = \frac{2}{3}$, so

$$Y \sim b\left(72, \frac{2}{3}\right), \quad E(Y) = np = 72 \cdot \frac{2}{3} = 48, \quad \text{Var}(Y) = np(1-p) = 72 \cdot \frac{2}{3} \cdot \frac{1}{3} = 16,$$

and the standard deviation is $\sqrt{16} = 4$. By the central limit theorem,

$$Y \dot{\sim} N(48, 4^2).$$

Step 3: Approximate $P(Y > 50)$.

$$\begin{aligned} P(Y > 50) &= P\left(\frac{Y - 48}{4} > \frac{50 - 48}{4}\right) = P(Z > 0.5) \\ &= 1 - \Phi(0.5) \approx 1 - 0.6915 = 0.3085, \end{aligned}$$

where $\Phi(\cdot)$ is the standard normal cdf.

With continuity correction. Since “more than 50” means $Y \geq 51$ for the integer-valued Y , we use the cutoff 50.5:

$$P(Y \geq 51) \approx P\left(Z > \frac{50.5 - 48}{4}\right) = P(Z > 0.625) = 1 - \Phi(0.625) \approx 0.266.$$

For comparison, the exact binomial probability is 0.269, so the continuity-corrected value (0.266) is much closer than the uncorrected one (0.3085).